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DEVELOPMENT ARTICLE

Validating student AI competency self-efficacy (SAICS) scale and its framework

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Abstract
Nurturing student artificial intelligence (AI) competency is crucial in the future of K-12 education. Students with strong AI competency should be able to ethically, safely, healthily, and productively integrate AI into their learning. Research on student AI competency is still in its infancy, primarily focusing on theoretical and professional discussions, along with qualitative investigations. This two-stage study aims to propose an AI competency framework for students and confirm the reliability and validity of its scale—student AI competency self-efficacy (SAICS)—in K-12 education. In stage 1, we used a three-round Delphi study to propose the framework and its scale. The framework has eight dimensions: interdisciplinary learning with AI, assessment with AI, decision-making with AI, data, ethics and AI, designing AI, multimedia creation with AI, human-centric learning, and confidence with AI. Each dimension contains four items. In stage 2, we involved 448 students to validate the scale using confirmatory factor analysis and model comparisons. The analyses showed that the scale is consistent across male and female students. The SAICS scale comprises 32 items and addresses eight dimensions of AI competency. Researchers can use the framework and SAICS to design their interventions and correlational research associated with student AI competency. Teachers can use them to develop learning outcomes for AI-based learning activities, and policymakers can use them to establish national AI standards.

Keywords AI competency · Scale development · K-12 education · AI literacy · Delphi study

Introduction

Artificial intelligence (AI) and education for all, moving from the professional to the mainstream, is a global initiative (Chiu, 2021; Touretzky et al., 2023). Understanding AI, including how it operates and the societal benefits it may provide, is the primary step toward a safe and healthy life in an AI-driven society. Most of us will live, learn, and work with AI, such as ChatGPT and Copilot, in the future at home, in schools, and in the workplace. Therefore, this education prepares youth for a future workforce that will require a greater level of AI literacy. A growing body of research suggests what AI literacy is and how to

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measure it (Chiu, 2025; Chiu et al., 2024; Kong et al., 2021; Laupichler et al., 2023; Long & Magerko, 2020; Zhou et al., 2025). Their findings have contributed to AI and education research and practice by providing insights into the expected learning outcomes and potential content for curricula. These studies focus on more mature learners or users, such as those in higher education (Kong et al., 2021; Long & Magerko, 2020) and non-AI experts (Laupichler et al., 2023). Their results were driven by the perspectives of university teaching staff, the opinions of researchers, and the literature review and might not be suitable for younger children, such as those in grades K–12, due to their developmental stage. Thus, more studies on defining and measuring AI for younger students from other participant perspectives, such as teachers, are needed.

AI literacy places greater emphasis on skills, but AI competency acknowledges a wider range of knowledge, capabilities, and attitudes (Chiu et al., 2024; UNESCO, 2024). As stated by Falloon (2020), the concept of competency needs ongoing adaptation to account for the progressive advancements in digital systems. This necessitates that individuals assess their present capabilities and needs and, if needed, review their learning to adapt to the dynamic digital landscape, possibilities, and challenges that are propelled by advancements in emerging technologies. Compared to non-AI technology, AI is a more disruptive and ever-evolving technology; engaging in updating its technical knowledge is required for student AI competency (Chiu, 2024; Chiu et al., 2024; Falloon, 2020; Liu et al., 2021; Zhou et al., 2025). Moreover, the current related studies mix the terms AI literacy, AI self-efficacy, and AI competency. More studies are needed to clarify these terms.

This study aims to propose an AI competency framework for K-12 students based on teacher perspectives, as well as to suggest and validate its scales, i.e., the student AI competency self-efficacy (SAICS) scale. The findings are expected to assist researchers in determining the effectiveness of empirical studies aimed at improving student AI competency, as well as teachers in assessing their students' competency to apply AI in learning. Policymakers can use our findings to establish their national standards for AI competency. Overall, the findings would help us understand the key components of K-12 student AI competency.

Conceptual background

Measuring AI literacy

AI literacy is defined as the ability to critically assess AI technology, interact and communicate with AI in an efficient manner, and use AI as a tool at home, at work, and online (Long & Magerko, 2020). Digital literacy refers to the capacity to utilize and assess digital information, tools, and services appropriately and to incorporate this knowledge into lifelong learning processes (Bawden, 2008) and is viewed as a prerequisite of AI literacy (Long & Magerko, 2020). Individuals should grasp how to utilize computers to make sense of AI. Moreover, computational literacy is defined as the ability to explore, communicate, and express ideas through code (diSessa, 2000) and is not always required for AI literacy (Chiu, 2021; Long & Magerko, 2020). Coding skills are necessary for developing but not using AI. Users could effectively, ethically, and safely use AI in their daily lives without mastering coding skills. In addition, data literacy refers to the capacity to comprehend, interpret, engage with, evaluate, and debate data as part of a larger process of inquiry into the world (D'Ignazio, 2017) and is overlapped with AI literacy (Chiu, 2021, 2025; Long

& Magerko, 2020). It is because the key idea of AI is learning from data. Overall, having a clear idea of what "AI literacy" is can help with mainstream education and skill assessments related to AI.

Literature suggests frameworks to capture constructs or items related to AI literacy and inform scholars to develop scales for AI education research. Long and Magerko (2020) used an interdisciplinary literature review approach to present one of the first comprehensive frameworks, consisting of five major dimensions and 17 areas for AI literacy assessment. The dimensions are: What is AI? What can AI do? How does AI work? How should AI be used? How do people perceive AI? The suggested areas are technical and too many, which are more appropriate for students in higher education or employees in the workplace. For example, the area "understanding that agents are programmable" is not for younger students. Laupichler et al. (2023) used a Delphi study to propose 38 self-reported items as an AI literacy scale. The scale has no dimensions, i.e., the whole set of items should be used for research or practice. This study is similar to Long and Magerko's (2020), and its items are too specific and technical. Kong et al. (2021) developed and used a 10-item AI literacy test (perceptions of their own levels of AI literacy) and a 17-item AI Empowerment Survey (sense of empowerment to deal with AI) to assess student AI learning. It did not present a whole set of instruments. Overall, the findings of these major studies have different views on how to measure AI literacy. All of them used self-reported items to assess participants' AI literacy. This self-reported approach was used in many related studies (Lin et al., 2021; Shih et al., 2021).

However, a self-reported approach can only examine participants' perceptions of using and evaluating AI (i.e., AI self-efficacy) but not their abilities and skills (i.e., AI literacy). AI literacy should be measured by tests or examinations similar to international literacy assessment programs such as the Program for International Student Assessment and Trends in International Mathematics and Science Studies. This argument is also strongly supported by the study of the AI self-efficacy scale done by Wang and Chuang (2023). The study developed and validated the self-efficacy scale consisting of four dimensions—assistance, anthropomorphic interaction, comfort with AI, and technological skills—using 317 adults. The format of the items (i.e., self-reported feeling, thinking) is similar to that in the studies discussed in the previous paragraph. Moreover, self-reported approaches have been studied in the majority of AI education studies on anxiety and attitude toward AI, as they focus more on how individuals feel and think. For example, AI attitude and AI for social good (Chiu et al., 2024), general attitudes towards AI (Schepman & Rodway, 2020), and AI anxiety (Wang & Wang, 2022). Given that we did not know what to include in AI education for the public or the mainstream, this is the most sensible course of action. Overall, both AI self-efficacy and AI literacy should be included in a more comprehensive assessment of student learning.

AI literacy and competency for school students

Self-efficacy concerns beliefs about their ability to know and apply; AI literacy concerns knowing, applying, and evaluating; and AI competency is about applying and evaluating the knowledge in a beneficial way (Falloon, 2020; Wang & Chuang, 2023; Zhou et al., 2025). These three concepts influence how to measure individuals' AI learning and understanding. In AI education empirical research, using self-efficacy is more popular. Two plausible explanations are (i) lack of reliable and validated instruments for AI literacy and competency and (ii) literacy and competency assessment depends on a research or teaching

context (e.g., learning content in an intervention should be aligned with the tests). Overall, using self-efficacy in assessing student AI competency would be more effective.

As we discussed, digital literacy is a prerequisite for digital competency (Long & Magerko, 2020). The move from AI literacy to competency is supported by various United Nations Educational, Scientific, and Cultural Organization (UNESCO) reports and Falloon's (2020) study on digital competency. Falloon (2020) clearly distinguished the difference between digital literacy and competency. He encouraged scholars and educators to have more comprehensive and broad-based conceptualizations than the current technical and literacy frameworks. It is because young people require ever-increasingly complex knowledge and abilities to operate in a variety of digitally mediated environments in an ethical, responsible, safe, and productive manner (Falloon, 2020). AI competency clearly involves more than understanding what AI is, how AI works, how to use AI, and what impact AI brings (Chiu et al., 2024). Given the fast and disruptive evolution of AI technology, this mindset is vital for AI learning. AI-literate users are able to leverage classic AI technologies in their daily lives in an effective and ethical manner, whereas AI-competent users are at ease with emerging AI technologies for specific purposes (Chiu, 2024; Chiu & Sanusi, 2024; Falloon, 2020; Liu et al., 2021; Zhou et al., 2025). For example, AI-literate students understand AI knowledge well in general, while AI-competent students can effectively use AI in their learning (project-based essay writing). To integrate AI into classrooms, it is advisable to embrace the concept of AI competency. The United Nations Educational, Scientific, and Cultural Organization (UNESCO) drafted an AI competency framework for K–12 students: "the knowledge, skills, and attitudes students should acquire to understand and actively engage with AI in a safe and ethical manner in school and beyond" (UNESCO, 2024). The framework consists of four primary components: a human-centric mindset, the ethics of AI, AI techniques and applications, and AI system design. Each component has three levels: understand, apply, and create. This framework is more about AI knowledge and understanding, focusing less on students' competency using AI in their learning. Chiu et al., (2024, 2025) took a different perspective and advocated that the definitions of AI competency should emphasize a specific context, i.e., K-12 students should be competent to use AI in their activities.

Teacher perspective for assessing younger student AI learning

AI competency is frequently employed to define the competency of non-engineering rather than engineering students (Laupichler et al., 2023; Long & Magerko, 2020; Wang & Chuang, 2023). Without obtaining formal professional training in AI, non-engineers are those who employ AI systems rather than build them. Non-engineering includes school students, who utilize AI applications for learning purposes. Therefore, AI and education for students from general education streams is therefore a global educational initiative (Chiu, 2024; UNESCO, 2024; Yau et al., 2022). Related studies on designing and creating AI learning activities for K–12 students were conducted worldwide in Europe, Korea, Hong Kong, China, and the United States. Teacher viewpoints were extensively included in the design of a number of these studies (Chiu, 2021; Kim et al., 2021; Williams et al., 2023; Yau et al., 2022). In comparison to academics, school teachers have a deeper understanding of younger students' learning and needs. However, most current studies on AI learning assessment are recommended by academics and focus on more senior learners or users. These may neglect teacher perspectives and sense-making (Chiu & Chai, 2020); however, teachers' beliefs and views will decide what AI learning and assessment look like. Hence,

it is important to get the views of school teachers when designing assessment approaches for younger students (K–12 in this study).

Research gaps and significance of this study

Our literature review provides three major arguments to support the significance of this study. First, measuring AI learning should go beyond skills and abilities (literacy) due to its rapidly emerging and disruptive nature (Chiu, 2024, 2025; Falloon, 2020; Liu et al., 2021). We suggest including more AI-based learning activities in the framework. Second, most current studies were designed for more mature participants, such as students in higher education and employees in the workplace, and took AI professional or academic perspectives (Laupichler et al., 2023; Long & Magerko, 2020; UNESCO, 2024; Wang & Chuang, 2023). Their findings were not appropriate for younger students and suggested that teachers' perspectives should be included in the research. Third, the three terms used for AI self-efficacy, literacy, and competency are mixed in the literature (Kong et al., 2021; Laupichler et al., 2023). Accordingly, this study significantly contributes to AI and education by clarifying AI competency by suggesting its assessment. Researchers and educators could use the findings to design their studies and evaluate their student learning in AI, respectively, to advance AI education research.

This study and method

Research goal

This study is to propose an AI competency framework for K-12 students from the point of view of school teachers and validate its measurement items. More specifically, we suggest the key components for student AI competency, followed by developing and validating a student artificial intelligence competency self-efficacy (SAICS) scale. Accordingly, the three main research questions are:

RQ1: What are the dimensions and their self-reported items for student AI competency from a K-12 teacher perspective?
RQ2: Is the SAICS scale quantitatively validated?
RQ3: Are there any significant differences in the SAICS scale based on student gender?

We used a two-stage approach to answer the three RQs. In stage 1, we used a Delphi study to identify the key component of student AI competency (RQ1) and develop items for SAICS. This stage is to construct a set of student AI competencies. The competencies were rated for their importance by a group of experienced teacher panelists. In stage 2, we used quantitative analyses to validate the SAICS scale (RQ2 and RQ3). The items measuring the competency were validated using confirmatory factor analysis (CFA).

Stage 1: a Delphi study

A Delphi study is a widely recognized methodology used to address research questions by identifying a consensus viewpoint among a group of specialists. It facilitates participant

introspection, enabling them to refine and review their opinions and ideas considering the anonymous perspectives expressed by others (Barrett & Heale, 2020). Three Delphi rounds are sufficient to establish an equilibrium where further rounds do not significantly alter the findings (Laupichler et al., 2023; Teixeira et al., 2020). Accordingly, we used a three-round process to answer the three research questions, which is supported by a similar study by Laupichler et al. (2023).

Every Delphi study has its own set of guidelines for attaining consensus on every topic (Barrett & Heale, 2020; Keeney et al., 2006). For example, it is easier to get all survey respondents to agree on three items than it is to get them to agree on twenty items. The process of developing agreement levels is intrinsically subjective and requires consideration of the questions posed, the range of possible options, and the size of the panel. According to the review study of Diamond et al. (2014), the agreement levels ranged from approximately 50–95%. This study used 75% as the agreement level, similar to the study of Teixeira et al. (2020). In each round, the participants gave their feedback on the items for AI literacy, self-efficacy, and self-reflective mindset. They may suggest removing or combining some of the items or dimensions and adding new items or dimensions. Moreover, the determination of sample size for Delphi research lacks unanimity; hence, no minimum panel size is established (Barrett & Heale, 2020; Chiu, 2024; Jorm, 2015). Prior studies employed ten to thirty-four experts as participants (Ahmadi et al., 2023; Barrett & Heale, 2020; Teixeira et al., 2020). The expert participants of this study were K–12 AI education professionals; that is, they possess extensive knowledge and expertise in developing AI learning activities for K–12 levels. To ensure that the participants are knowledgeable, the selection criteria are holding undergraduate or graduate degrees in computer science, technology, STEM, or engineering; (ii) having at least 3 years of formal experience teaching AI (classroom teaching); and (iii) having completed at least twelve hours of professional development on AI education. With the understanding that K–12 AI education is still in its infancy and that most related research started in 2019, having taught for 3 years is considered experienced.

This Delphi study was conducted in Hong Kong, which started the first AI K–12 education project in August 2019. In Hong Kong, an official AI curriculum for seventh through ninth graders was developed in September 2023. Hong Kong schools taught AI using the learning materials and evaluation exam.

We invited 40 teachers from a professional development program on AI for education to take part, but only 29 teachers met the requirements and agreed. They all used the official curriculum resources for their AI classes. Twenty-two of them were male, which represented the gender ratios of AI teachers (representative sampling). The average AI teaching experience is 2.8 years old, and the average age is 31 years old. The main research team included four AI and education scholars from different regions—Hong Kong, Africa, Finland, and Turkey—and three teacher-researchers on AI education.

This stage intentionally developed key components of K-12 student competency and their items. First, to develop the initial materials for the Delphi Round 1, the research team compiled the items by reviewing the related studies. For the first set, the reviewed studies included the official curriculum and the studies of UNESCO (2024), Chiu et al. (2024), Long and Magerko (2020), Wang and Chuang (2023), and Wu and Chen (2017).

Figure 1 shows the process of our Delphi study. In Round 1, we distributed the lists to the teacher participants. The teachers gave qualitative comments on the dimensions and items. They also assessed if the items reflect student AI learning at the K-12 level. The participants were asked to revise, remove, combine, and add new items. After each round, the research team consolidated the participants' feedback to revise the dimensions and items. Items with substantial changes are considered new items in the next round. The items that

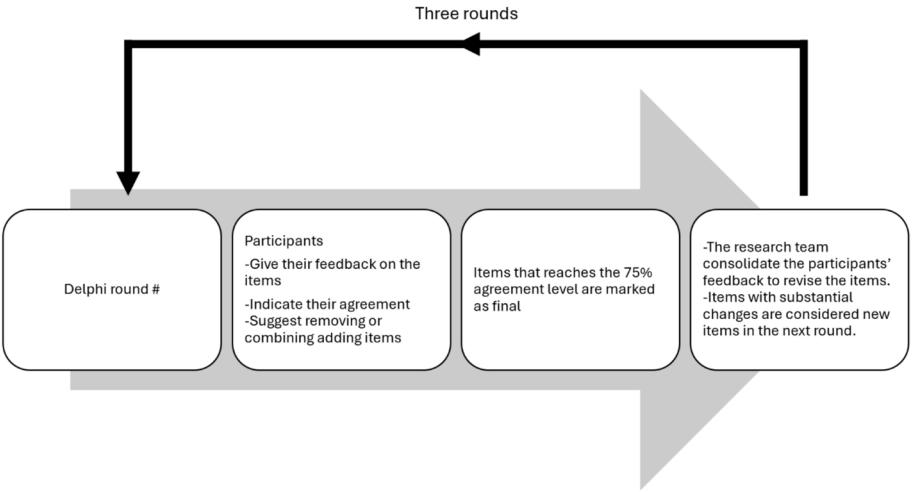

Fig. 1 The process of our Delphi study

reached the agreement level were marked as final. In Rounds 2 and 3, the teachers received the most updated lists of activities, together with information on the agreement level. They continued to give comments on the dimensions and items.

Stage 2: quantitative analyses to validate SAICS scale

Four hundred and forty-eight student participants from six Hong Kong secondary schools (Grades 7–12) from various socioeconomic backgrounds. They aged from 12 to 17 years old (M = 14.8, SD = 2.8; male: 222 and female: 226). The schools were chosen from three school-university partnership programs overseen by the corresponding author. The items used in this stage are the results from the Delphi student and were assessed on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree).

To respond to RQ2, the measurement model was evaluated using traditional fit indices and standards, such as Kline (2015), which include the chi-square tests of fit, the comparative fit index (CFI), and the root mean square error of approximation (RMSEA) with a 90% confidence interval. To address RQ3, we used changes in CFI as indicated for comparing models to see if the TAICS scale is consistent across gender groupings (Zhan, 2022).

Results

Stage 1

Initial materials

Table 1 shows the draft for the research team's discussions. The draft was developed based on Chiu et al. (2024), Long and Magerko (2020), UNESCO (2024), Wang and Chuang (2023), and Wu and Chen (2017). The research team wrote the initial draft by modifying

Table 1 The first draft for the panel's discussion

Items	Dimension
First draft for the panel's discussion	
• Some AI technologies/products make learning easier • I find that AI technologies/products are helpful for learning • AI technologies/products are good aids for learning • Using AI technologies/products makes learning more interesting • I'm confident in my ability to learn simple programming of AI technologies/products if I were provided the necessary tools • AI technologies/products help me to save a lot of time • I find it easy to get AI technologies/products to do what I want them to do	Assistance
• I think the interactive process of AI technologies/products is very vivid, just like chatting with a real person • I think the way that AI technologies/products express content when interacting is unique, just like a real person • I think there is no difference between the dialogue method of AI technologies/products compared with the dialogue with real people • I think the tone of AI technologies/products when interacting is the same as that of real people • I feel that the way of expression of AI technologies/products in the interactive text is the same as that of real people	Anthropomorphic interaction
• When interacting with AI technologies/products, I feel very calm • When interacting with AI technologies/products, I find it easy • When interacting with AI technologies/products, I feel comfortable in my heart • When interacting with AI technologies/products, I feel very peaceful • When interacting with AI technologies/products, I feel very relaxed • I can happily interact with AI technologies/products smoothly	Comfort with AI
• When using AI technologies/products, I am not worried that I might press the wrong button and cause risks • When using AI technologies/products, I am not worried that I might press the wrong button and damage it • When using AI technologies/products, there is nothing that I do not know why • AI technologies/products jargon does not baffle me	Technological skills
Self-reflective mindset	
• I frequently examine my feelings • I frequently take time to reflect on my thoughts • I often think about the way I feel about things • It is important for me to evaluate the things that I do • I am very interested in examining what I think about • It is important to me to try to understand what my feelings mean • I have a definite need to understand the way that my mind works • It is important to me to be able to understand how my thoughts arise	Engagement in self-reflection Needs for self-reflection

the four dimensions and 22 items of Wang and Chuang (2023) to develop initial material. We revised the wording to fit the younger student and made the sentence structure more consistent. For example, most of the revised items started with "I…" or "It is…". We changed "assistance" into "support for learning" and "anthropomorphic interaction" into "human-like interaction". We also added perceived ease of use (Wu & Chen, 2017) because they are strongly related to self-efficacy. Moreover, we modified eight items from the study of Grant et al. (2002) to develop the items for assessing self-reflective mindset for the AI questionnaire. We also add five more dimensions, such as assessment with AI,

creation with AI, critical thinking, perceived ease to use as well as data, ethics, and AI. The initial materials have 10 dimensions and 50 items for Delphi Round 1, see Table 2.

Three Delphi rounds

Three Delphi rounds were completed by all the participants. Table 3 shows the results of Rounds 1, 2, and 3. Eight dimensions were revised and finalized from 10 dimensions in the initial lists. The labels of two dimensions remain the same, and they are "Assessment with AI (ASAI)" and "Data, ethics, and AI (DEAI)". Five of them were renamed: "Interdisciplinary learning with AI (ILAI)" instead of "Support for learning"; "Multimedia creation with AI (MCAI)" instead of "Creation with AI'; "Decision making with AI (DMAI)" instead of "Critical thinking"; "Human-centric learning (HCLG)" instead of "Human-like interaction"; and Designing AI (DGAI)" instead of "Technological skills". Three of them were integrated into one dimension: "Confidence with AI (CFAI)" instead of "Comfort with AI", "Perceived ease to use", and "Engagement in self-reflection". Each of the dimensions has 4 items. Therefore, the final list has eight dimensions and 32 items (see the ✓ in Table 3).

Stage 2

Internal consistency, variability and validity of SAICS scale

As shown in Table 4, TAISCS scores were internally consistent for all eight dimensions—ILAI, ASAI, DMAI, DEAI, DGAI, MCAI, HCLG, and CFAI. All Cronbach's alpha (CR) values are equal to or greater than 0.89; All dimensions were near to normal distributions (skewness and kurtosis between −1 and 1) and had sufficient variability (range and standard deviation); All dimensions had high factor loadings, ranging from 0.76 to 0.96 (> 0.80). The correlation matrix between SAICS scale scores supported the scale's convergent validity, with all correlated dimensions ($r > 0.34$; $p < .05$).

CFA

In the CFA, the fitness indices of the measured items revealed a good model fit regarding the measurement model's goodness-of-fit: $x^2/df = 2.39$ (< 5.0); RMSEA = 0.056 (< 0.08); SRMR = 0.03 (< 0.05); PNFI = 0.81 (> 0.50); CFI = 0.95 (> 0.90) (Kline, 2015; Zhan, 2022). Figure 2 shows the path relations and coefficients between the items. These results show that the measurement model of SAICS scale has high validity and reliability scores (RQ2).

Gender differences across SAICS scale dimensions

We examined the gender differences in SAICS (RQ3). A multiple-group CFA was performed to determine whether the measurement models differed significantly between gender (male vs. female) groups. A drop of 0.01 in CFI is regarded as suitable for detecting lack of invariance across several groups in structural equation modeling (Cheung & Rensvold, 2002); thus, changes in CFI are advised for comparing models with sufficiently large sample sizes (Zhan, 2022).

Table 2 The initial materials for Round 1

Items	Initial list
• I know how to use AI technology to support my learning • I know how to use AI technology for self-reflection in learning • I know how to use AI technology to solve my learning difficulties • I know how to choose the right AI technology to support my learning • I find learning with AI technology more effective than without • I find that AI technology is helpful for learning • I believe AI technology can help me learn a topic by myself • I believe AI technology can help me find additional questions for my learning	Support for learning
• I can use AI to check whether my solutions are correct • I can use AI to generate questions for further learning • I can use AI to get feedback to write essays	Assessment with AI
• I can use AI to create images for my learning (e.g., presentations, communications, assignments) • I can use AI to create videos for my learning • I can use AI to create audio for my learning	Creation with AI
• I have good critical thinking to learn with AI • I am able tell whether AI can benefit my learning • I could judge if an essay generated by AI is correct for my learning • I can judge if an image/video essay generated by AI is correct for my learning	Critical thinking
• I think the interactive process of AI technology is very vivid, just like chatting with a real person • I think the way that AI technology expresses content when interacting is unique, just like a real person • I think there is no difference between the method of AI technology compared with the dialogue with real people • I think the tone of AI technology when interacting is the same as that of real people • I feel that the way of expression of AI technology in the interactive text is the same as that of real people	Human-like interaction
• I feel comfortable using AI technology in learning • I feel confident using AI technology in learning • I feel relaxed using AI technology in learning • I feel calm using AI technology in learning • I feel peaceful using AI technology in learning • I can happily interact with AI technology smoothly	Comfort with AI
• When using AI technology, I am not worried that I might press the wrong button and cause risks • When using AI technology, I am not worried that I might press the wrong button and damage it • When using AI technology, there is nothing that I do not know why • AI technologies/products jargon does not baffle me	Technological skills
• I find it easy to get AI technology to do what I want it to do • I find AI technology helps me save a lot of time • I find AI technology easy to use for learning • I find it easy to learn AI technology • I believe it is easy to become proficient in using AI for learning • I believe the interaction with AI technology is clear and understandable	Perceived ease to use
• I know how to protect my personal data when using AI • I understand that AI technology sometimes gives a biased output • I understand that AI technology sometimes generates incorrect output • I understand that AI technology can generate fake information	Data, ethics, and AI

Table 2 (continued)

Items	Initial list
• I frequently examine AI knowledge • I frequently take time to reflect on my AI knowledge • I often think about the way I use AI in learning • It is important for me to evaluate the AI technology that I use • I am very interested in examining how I use AI technology in learning • It is important to me to try to understand the advantages and risks of the AI technology I use • I have a definite need to understand how AI technology processes data	Engagement in self-reflection

Table 5 shows the results of a multiple-group CFA to determine whether the structure of the SAICS scale was consistent between male and female teachers. The results showed that the baseline model is a good fit, with RMSEA = 0.057, PNFI = 0.761, and CFI = 0.916. The invariant measurement weight model provides a good fit, with RMSEA = 0.056, PNFI = 0.781, and CFI = 0.916. Similarly, the invariant structural covariances model fits well, with RMSEA = 0.056, PNFI = 0.806, and CFI = 0.911. Thus, all the models are suitable for comparison. The model comparison analysis revealed that both models of invariant measurement weight (change in CFI = 0.000) and invariant structural covariances (change in CFI = 0.005) differed by less than 0.01 from the baseline model. These results demonstrate that the measuring model is consistent across male and female students.

Discussion

This two-stage study seeks AI-experienced professional teachers' opinions to create an AI competency framework for students and validate its SAICS scale. The results offer two major implications: an AI competency framework for students and a validated SAICS scale (see Table 6).

Eight dimensions in AI competency framework for students

The results of the Delphi study suggest eight dimensions in an AI competency framework for students. They are Interdisciplinary Learning with AI (ILAI), Assessment with AI (ASAI), Decision Making with AI (DMAI), Data, Ethics, and AI (DEAI), Designing AI (DGAI), Multimedia Creation with AI (MCAI), Human-Centric Learning (HCLG), and Confidence with AI (CFAI). The framework focuses on student capacity to use AI to support their learning ethically, healthily, and productively. DMAI, DEAI, DGAI, and HCLG are UNESCO's (2024) AI competency framework for school students, and Chiu et al. (2024); ILAI, ASAI, MCAI, and CFAI are aligned with the studies of Chiu et al. (2024), Chiu (2024), Lin et al. (2021), Shih et al. (2021), and Wang and Chuang (2023). This framework considers the impact of generative AI technologies such as ChatGPT and other educational chatbots and includes interdisciplinary learning (ILAI), assessment (ASAI), and multimedia creation (HCAI), including text-to-image and text-to-video (Chiu, 2024; Chiu et al., 2024; Zhang & Zhang, 2024). It also emphasizes the importance of meaningful learning, remembering, and understanding when using AI in learning (HCLG) (Guo, 2024). Students are required to make their judgments on the outputs created

Table 3 The first list and the results of Round 1, 2, and 3 in the self-efficacy questionnaire (self-reported items)

Items for student AI competency				
Items	Initial list	Round 1	Round 2	Round 3
• I know how to use AI technology to support my learning	Interdisciplinary learning with AI (Support for learning)	@		
• I know how to use AI technology for self-reflection in learning		@		
• I know how to use AI technology to solve my learning difficulties		@		
• I know how to choose the right AI technology to support my learning		@		
• I find learning with AI technology more effective than without			@	
• I find that AI technology is helpful for learning		@	@	
• I believe AI technology can help me learn a topic by myself			@	
• I believe AI technology can help me find additional questions for my learning				
✓ I can use AI to get different alternative information to solve real-life problems better			+	#
✓ I can use AI to gain knowledge in a discipline I am unfamiliar with			+	#
✓ I can use AI to conduct inquiry-based learning in other disciplines			+	#
✓ I can use AI to find relevant information outside the classroom curriculum quickly and efficiently			+	#
✓ I can use AI to check whether my solutions are correct	Assessment with AI	#	#	#
✓ I can use AI to generate questions for further learning		#	#	#
• I can use AI to get feedback to write essays		^		
✓ I can use AI to get feedback to improve my ideas		^	#	#
✓ I can use AI to get ideas to complete my learning tasks			+	#
✓ I can use AI to create images for my learning (e.g., presentations, communications, assignments)	Multimedia creation with AI (Creation with AI)	#	#	#
✓ I can use AI to create videos for my learning		#	#	#
✓ I can use AI to create audio for my learning		#	#	#
✓ I can use AI to create slides for my learning		+	#	#

Table 3 (continued)

Items for student AI competency				
Items	Initial list	Round 1	Round 2	Round 3
• I have good critical thinking to learn with AI	Decision making with AI (Critical thinking)	^		
• I am able tell whether AI can benefit my learning		^	#	#
✓ I can ask for more resources based on the recommendations by AI for my learning		^	#	#
✓ I can judge if resources (e.g., websites, videos) recommended by AI are appropriate for my learning		^		#
✓ I can judge if an essay generated by AI is correct for my learning		#		#
✓ I can judge if an image/video essay generated by AI is correct for my learning		#		
• I think the interactive process of AI technology is very vivid, just like chatting with a real person	Human-centric learning (Human-like interaction)	@		
• I think the way that AI technology expresses content when interacting is unique, just like a real person		@		
• I think there is no difference between the method of AI technology compared with the dialogue with real people		@		
• I think the tone of AI technology when interacting is the same as that of real people		@		
• I feel that the way of expression of AI technology in the interactive text is the same as that of real people		@		
✓ I believe that AI is a tool to enhance my learning experience, rather than just a means to finish assignments		+	#	#
✓ I can leverage AI to boost my understanding and skills, not merely to complete tasks		+	#	#
✓ I can ensure my essays are original when co-writing with AI			+	#
✓ I can collaborate with AI to create slides that are my ideas			+	#

Table 3 (continued)

Items for student AI competency

Items	Initial list	Round 1	Round 2	Round 3
• When using AI technology, I am not worried that I might press the wrong button and cause risks	Designing AI (Technological skills)	@		
• When using AI technology, I am not worried that I might press the wrong button and damage it		@		
• When using AI technology, there is nothing that I do not know why		@		
• AI technologies/products jargon does not baffle me		@		
✓ I know how to collect data for designing machine learning		+	#	#
✓ I know how to create a machine learning application		+	#	#
✓ I know how to categorize data for designing machine learning		+	#	#
✓ I know how to evaluate my created machine learning		+	#	#
✓ I know how to protect my personal data when using AI	Data, ethics, and AI	#	#	#
✓ I understand that AI technology sometimes gives a biased output		#	#	#
✓ I understand that AI technology sometimes generates incorrect output		#	#	#
• I understand that AI technology can generate fake information		^		
✓ I understand that I should not create fake information		^	#	#
✓ I feel comfortable using AI technology in learning	Confidence with AI (Comfort with AI)	^	#	#
✓ I feel confident using AI technology in learning		^	#	#
• I feel relaxed using AI technology in learning		^		
• I feel calm using AI technology in learning		^		
• I feel peaceful using AI technology in learning		–		
• I can happily interact with AI technology smoothly		–		
✓ I am confident I will find new AI technology for learning		^	+	#
✓ I find it easy to learn AI technology		^	#	#

Table 3 (continued)

Items for student AI competency				
Items	Initial list	Round 1	Round 2	Round 3
• I find it easy to get AI technology to do what I want it to do	(Perceived ease to use)	^		
• I find AI technology helps me save a lot of time		–		
• I find AI technology easy to use for learning		–		
• I find it easy to learn AI technology		^		
• I believe it is easy to become proficient in using AI for learning		^		
• I believe the interaction with AI technology is clear and understandable		–		
• I frequently examine AI knowledge	(Engagement in self-reflection)	–		
• I frequently take time to reflect on my AI knowledge		–		
• I often think about the way I use AI in learning		–		
• It is important for me to evaluate the AI technology that I use		–		
• I am very interested in examining how I use AI technology in learning		^		
• It is important to me to try to understand the advantages and risks of the AI technology I use		–		
• I have a definite need to understand how AI technology processes data		–		

#Reach agreement level; @ removed; ^ combined or revised; + added

✓Indicates the final list that reaches agreement level

Table 4 Descriptive statistics and correlation among all the items in measurement model

Dimension	CR	Mean (SD)	1	2	3	4	5	6	7	8
1. ILAI	.91	3.21 (1.09)	–							
2. ASAI	.90	3.14 (1.09)	.54*	–						
3. DMAI	.89	3.18 (1.05)	.42*	.51*	–					
4. DEAI	.89	3.10 (1.03)	.52*	.50*	.49*	–				
5. DGAI	.91	3.03 (1.08)	.38*	.46*	.43*	.44*	–			
6. MCAI	.90	2.98 (1.12)	.50*	.48*	.45*	.55*	.47*	–		
7. HCLG	.96	3.07 (1.11)	.50*	.41*	.34*	.43*	.36*	.49*	–	
8. CFAI	.96	3.08 (1.05)	.46*	.35*	.40*	.40*	.35*	.41*	.50	–

*$p < .05$

or recommended by AI (DMAI). Designing AI applications (DGAI) and comprehending data and AI ethics (DEAI) can assist students in making informed learning decisions. This framework also emphasizes the emotional side; students should be confident in integrating AI into their learning (CFAI).

SAICS scale

Our findings showed that the SAICS scale, which we developed using a Delphi study in Stage 1, was reliable and valid. The SAICS scale was consistent among both female and male students. Our analyses demonstrated that the proposed eight-dimensional scale can measure student AI competency (see Table 3). The following provides detailed descriptions for each dimension.

- ILAI: This competency to use AI to support interdisciplinary learning (Chiu, 2024). AI can be a powerful tool for solving real-life problems by providing alternative information and perspectives. It helps in gaining knowledge in students' unfamiliar disciplines by offering tailored resources and explanations. AI supports inquiry-based learning across various fields, enabling deeper exploration and understanding. AI can quickly and efficiently find relevant information outside the classroom curriculum, making it easier to stay informed and up to date. Students should be able to enhance their problem-solving skills, broaden their knowledge base, and conduct thorough research in a more efficient manner, making AI an invaluable asset in interdisciplinary learning.
- ASAI: This competency is used to support assessment (Chiu, 2024; Weng et al., 2024; Xia et al., 2024). Students should be able to use AI to verify the correctness of their solutions, ensuring they are on the right track. AI can generate questions to further their learning, helping them delve deeper into subjects. They can use AI to get feedback to refine and improve their ideas. They can use AI to get creative suggestions to complete their learning tasks, making their learning process related to revision more efficient and effective. By leveraging AI, students can enhance their understanding, receive personalized feedback, and explore unfamiliar knowledge, ultimately enriching their learning experience.
- MCAI: This competency to use AI to create multimedia content (Zhang & Zhang, 2024). AI can enhance student learning by creating images for presentations, communications, and assignments. It can also generate videos and audio to support their

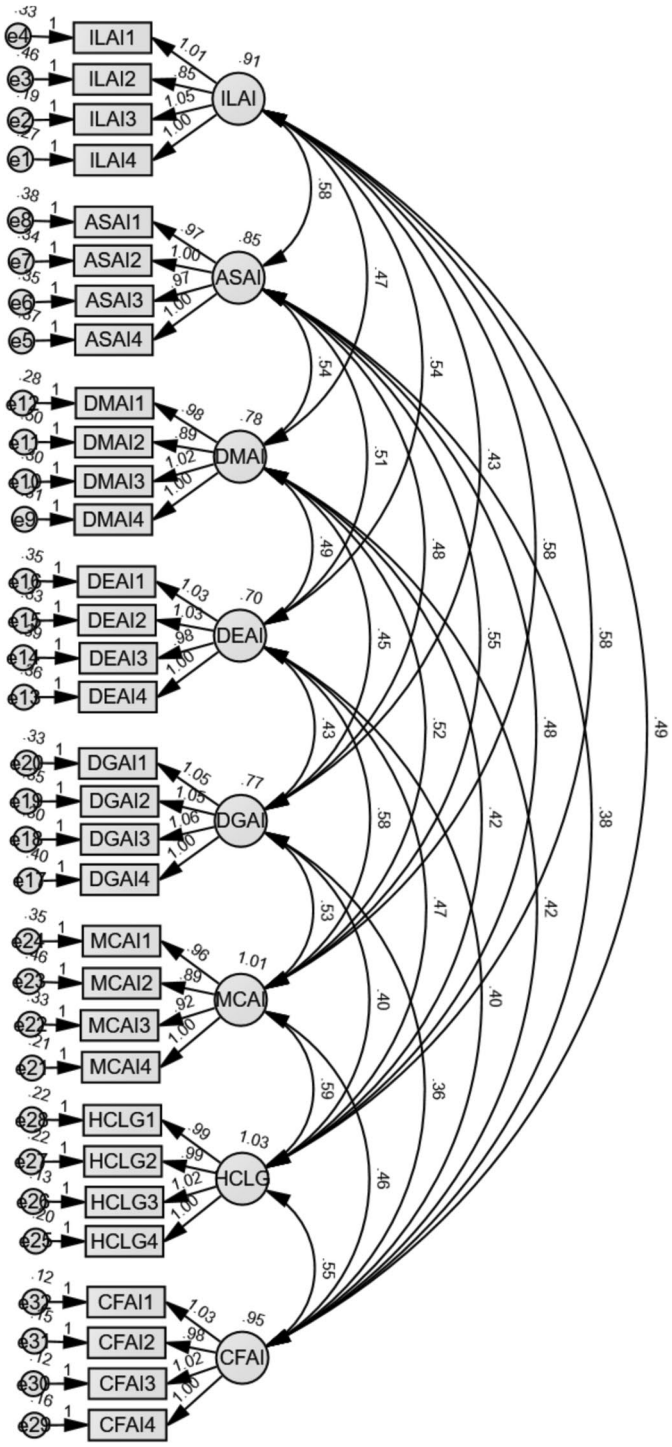

Fig. 2 Confirmatory factor analysis of 32 items of the SAICS scale

Table 5 Invariance tests across teachers of different genders

Model	x^2/df	RMSEA	PNFI	CFI	Change in CFI	Change in df
	Invariance across male and female teachers					
Baseline model	2.425	.057	.761	.916		
Invariant measurement weight model	2.382	.056	.781	.916	.000	28
Invariant structural covariances model	2.417	.056	.806	.911	.005	60

Table 6 The final SAICS scale

Dimension	Items
Interdisciplinary learning with AI	✓ I can use AI to get different alternative information to solve real-life problems better ✓ I can use AI to gain knowledge in a discipline I am unfamiliar with ✓ I can use AI to conduct inquiry-based learning in other disciplines ✓ I can use AI to find relevant information outside the classroom curriculum quickly and efficiently
Assessment with AI	✓ I can use AI to check whether my solutions are correct ✓ I can use AI to generate questions for further learning ✓ I can use AI to get feedback to improve my ideas ✓ I can use AI to get ideas to complete my learning tasks
Multimedia creation with AI	✓ I can use AI to create images for my learning (e.g., presentations, communications, assignments) ✓ I can use AI to create videos for my learning ✓ I can use AI to create audio for my learning ✓ I can use AI to create slides for my learning
Decision making with AI	✓ I can ask for more resources based on the recommendations by AI for my learning ✓ I can judge if resources (e.g., websites, videos) recommended by AI are appropriate for my learning ✓ I can judge if an essay generated by AI is correct for my learning ✓ I can judge if an image/video essay generated by AI is correct for my learning
Human-centric learning	✓ I believe that AI is a tool to enhance my learning experience, rather than just a means to finish assignments ✓ I can leverage AI to boost my understanding and skills, not merely to complete tasks ✓ I can ensure my essays are original when co-writing with AI ✓ I can collaborate with AI to create slides that are my ideas
Designing AI	✓ I know how to collect data for designing machine learning ✓ I know how to create a machine learning application ✓ I know how to categorize data for designing machine learning ✓ I know how to evaluate my created machine learning
Data, ethics, and AI	✓ I know how to protect my personal data when using AI ✓ I understand that AI technology sometimes gives a biased output ✓ I understand that AI technology sometimes generates incorrect output ✓ I understand that I should not create fake information
Confidence with AI	✓ I feel comfortable using AI technology in learning ✓ I feel confident using AI technology in learning ✓ I am confident I will find new AI technology for learning ✓ I find it easy to learn AI technology

learning tasks, making complex concepts more accessible. AI can help students create engaging slides for their learning tasks, ensuring their presentations are visually appealing and informative. Therefore, students should be able to produce high-quality, multimedia content that enriches their learning experience.

- DMAI: This competency is to make the right decisions on AI outputs and recommendations for learning. Students should be able to evaluate whether the suggested websites, videos, and other materials are suitable for their learning needs. They can assess the accuracy and relevance of AI-generated essays and images/videos for their needs. They should be able to ensure that the content they use is appropriate and beneficial for their learning goals.
- HCLG: This competency can be seen as awareness of using AI for meaningful learning, remembering, and understanding. Students should understand that AI is a powerful tool to enhance their learning experience, rather than just a means to finish assignments. They should use AI to boost their understanding and skills, not merely complete tasks. For example, when co-writing with AI, students ensure their essays remain original and reflect their own thoughts. Students can collaborate with AI to create slides that embody their ideas, making their presentations more engaging and personalized.
- DGAI: This competency is to design AI applications. Students should be proficient in creating machine learning applications, which allows them to develop innovative solutions. They should understand the process of collecting data for machine learning, ensuring the data is relevant and accurate. They should also be able to evaluate their created machine learning models, assessing their performance and making necessary improvements. This competency will help students know how to choose the right AI technology to support their learning, ensuring it aligns with their learning goals.
- DEAI: This competency is related to student understanding of data and AI ethics. Students should know how to protect their personal data when using AI, ensuring their privacy and security. They should understand that AI technology can sometimes produce biased or incorrect outputs and should be aware of the potential for AI to generate fake information. Recognizing these limitations, they are committed to using AI responsibly and ethically and understand the importance of not creating or spreading fake information. By being vigilant and informed, students can leverage AI effectively while maintaining integrity and accuracy in their work. This awareness helps students navigate the complexities of AI technology with confidence and responsibility.
- CFAI: This competency concerns student confidence to use AI in their learning. Students should feel comfortable and confident using AI technology in their learning. They should find it simple to learn and adapt to new AI tools, making the integration of AI into their learning seamless and beneficial. This confidence and ease with AI technology empower students to explore and leverage its full potential in their learning.

SAICS scale serves as a reference for research and practices

Student AI competency is linked to the successful integration of AI in education (Chiu, 2024; Chiu et al., 2024; UNESCO, 2024; Zhou et al., 2025). Students should be able to ethically, safely, healthily, and productively use AI to support their learning. The eight-dimension student AI competency and SAICS scale in this study allow K-12 education researchers and practitioners to methodically analyze student AI competency. They can be applied in the following proposed areas, among others.

- Researchers can use SAICS to reproduce and update the instruments used to evaluate their intervention and learning design for AI instruction in schools.
- Teacher educators can introduce this framework and SAICS in their teacher education program, as fostering student AI competency is also one of the main responsibilities of teachers.
- Teachers can utilize the framework and SAICS as learning objectives/outcomes to design AI learning activities. This is because assessment can influence learning design.
- Policymakers can leverage the framework and SAICS to create national AI competency standards, inform school and public education programs, and design assessment and policy parameters to ensure effective AI integration in education.

Conclusion, limitations and future research suggestions

This study proposes a student AI competency framework and its SAICS scale. It is crucial to recognize that AI education research is still in its infancy. The findings allow us to continue our work in this field. Moreover, AI technologies are projected to be integrated into K-12 education. The findings are critical in helping us understand what student AI competency should be and how to measure it. Moreover, future research should address the four limitations of this study when using and improving the framework and the SAICS scale, see the following.

- AI competency is linked to current literacies such as data, mathematics, science, and computation. This study focuses mostly on the level of AI competency. Future research should seek to provide a comprehensive picture by incorporating all relevant literacy factors.
- The SAICS scale is reliable and valid but lacks specificity across disciplines, focusing solely on assessing student competency in learning. There might be variations in different disciplines. Future research on AI competency in learning specific subjects is necessary.
- The SAICS scale's gender differences were the sole focus of this study. Future research should assess the impact of additional factors, including culture and socioeconomic background, on the SAICS scale.
- This scale may not be appropriate for a non-Chinese population, as the participants were Chinese students. Future research should adapt the language to the specific environment and be validated in a variety of communities, including non-Chinese or multiracial groups in regions such as the Middle East, South Asia, the United States, Europe, the United Kingdom, and Australia.

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Data availability The datasets used for the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests There is no conflict of interests between the author and participants.

Ethical approval This study got ethical clearance from the author's university.

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